Reliability of Power Electronics Converters for Solar Photovoltaic Applications

Edited by
Ahteshamul Haque, Frede Blaabjerg, Huai Wang, Yongheng Yang, Zainul Addin Jaffery

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About the Editors

Frede Blaabjerg is a full professor at Aalborg University's Centre of Reliable Power Electronics, Denmark. His current research interests include power electronics and its applications such as in wind turbines, PV systems, reliability, harmonics, and adjustable speed drives. He has authored or co-authored more than 600 journal papers, and co-authored or edited fourteen books in power electronics. He is recipient of 32 IEEE Prize Paper Awards, the IEEE PELS Distinguished Service Award in 2009, the IEEE William E. Newell Power Electronics Award 2014, Global Energy Prize in 2019 and the 2020 IEEE Edison Medal. He has been President of the IEEE Power Electronics Society in 2019-2020. He was nominated in 2014-2020 by Thomson Reuters as one of the 250 most cited researchers in engineering in the world.

Ahteshamul Haque is an assistant professor at Jamia Millia Islamia University, New Delhi, India. His research focuses on power electronics and its application in renewable energy, drives, and other areas. Prior to Jamia Millia Islamia, he was working for a multinational organisation. He has received patents and awards for his work, and established an Advanced Power Electronics Research Lab.

Huai Wang is a full professor at the Centre of Reliable Power Electronics (CORPE), Aalborg University, Denmark. His research addresses the fundamental challenges in modeling power electronic component failure mechanisms and application issues in system-level predictability, condition monitoring, circuit architecture, and robustness design. He was previously a visiting scientist at the ETH Zurich, Switzerland, the Massachusetts Institute of Technology, USA, and with the ABB Corporate Research Centre. His awards include the 2016 Richard M. Bass Outstanding Young Power Electronics Engineer Award and the 2014 Green Talents Award from the German Federal Ministry of Education and Research.

Zainul Abdin Jaffery is a professor and Head of the Department of Electrical Engineering, Jamia Millia Islamia University, New Delhi, India. He has published about 80 research papers in the area of Electronics and Electrical engineering in journals and conferences. His research focuses on digital signal processing, digital image processing, and their applications in power engineering and electronics engineering. He is a senior member of IEEE (USA).

Yongheng Yang is a ZJU100 Professor at Zhejiang University, China. He received the Ph.D. degree from Aalborg University in 2014, where he was an associate professor in 2018-2020. He has published more than 250 scientific papers and two monographs. He received the 2018 IET Renewable Power Generation Premium Award. He was the IEEE Denmark Section Chair during 2019-2020. He is an associate editor for several IET/IEEE journals.

Chapter 1

Power electronics converters for solar PV applications

Ahteshamul Haque[1]

1.1 Introduction

The demand for renewable energy-based power plants is growing exponentially worldwide due to the climate change threat. The other reason for this demand is the exponential growth in electrical energy demand for industrialization. The availability of conventional fossil fuel reservoirs such as coal is very limited. Solar photovoltaic (PV)-based power plant is the most acceptable among all renewable energy sources as the sunlight is relatively available in abundance in most of the regions.

The total solar PV installed capacity is thus increasing worldwide, and it is forecasted that growth will continue, as shown in Figures 1.1 and 1.2, [2].

Almost in every business sector, the world is facing a growing recession, but in solar PV-based plant area, the job prospects have increased significantly, as indicated in Figure 1.3, and many opportunities are created at various levels, which helps in solving unemployment issues worldwide.

Based on the above data, it can be concluded that solar energy is the fastest growing renewable energy generation technique and most of the researchers are focusing on the development of a highly efficient and lossless method to achieve maximum power possible. Moreover, solar PV plants have technological challenges i.e. power conversion, quality factor, and safety issues. To address these issues various technological standards are made by regulating agencies, e.g., power electronics converters, control strategy, low-voltage ride-through (LVRT) [3, 4], anti-islanding [5, 6], and reliability, which will be discussed in the coming sections and chapters in detail.

[1]Advance Power Electronics Research Laboratory, Department of Electrical Engineering, Jamia Millia Islamia (A Central University), New Delhi, India

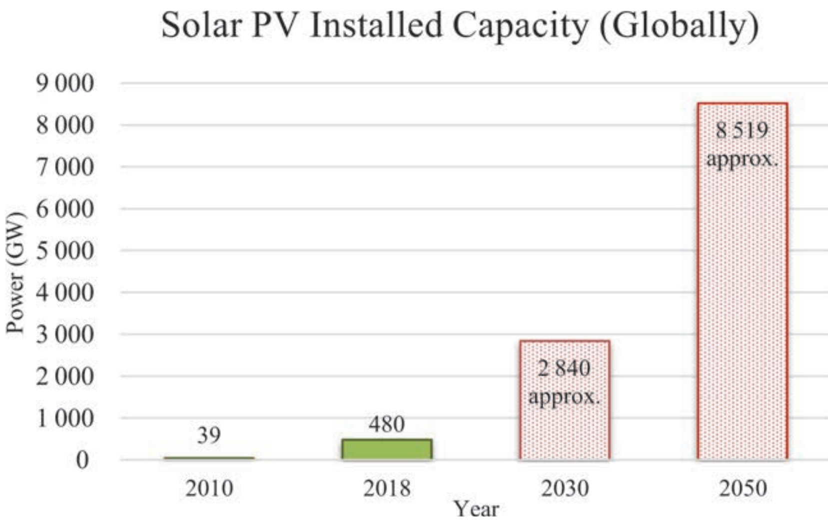

Figure 1.1 Solar PV installed capacity at world level [1]

1.2 Role of power electronics in solar PV systems

The layout of a solar PV power plant is shown in Figure 1.4. The solar PV output is DC and variable, i.e., it depends on ambient conditions (temperature, solar irradiance, etc.). These variable electrical signals need to be regulated as per the desired shape and magnitude. Power electronics converters are used to regulate the PV output. Moreover, power electronics converters are also used to convert the DC voltage into the AC voltage of the desired magnitude and frequency. The ability of

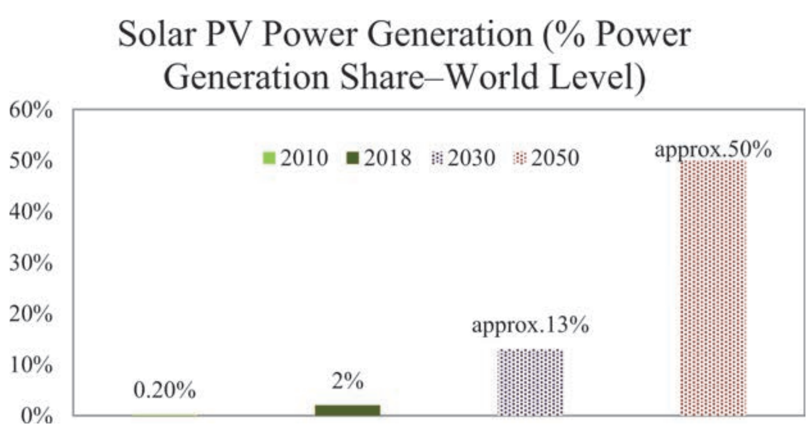

Figure 1.2 Power generation share of solar PV-based plants at world level [1]

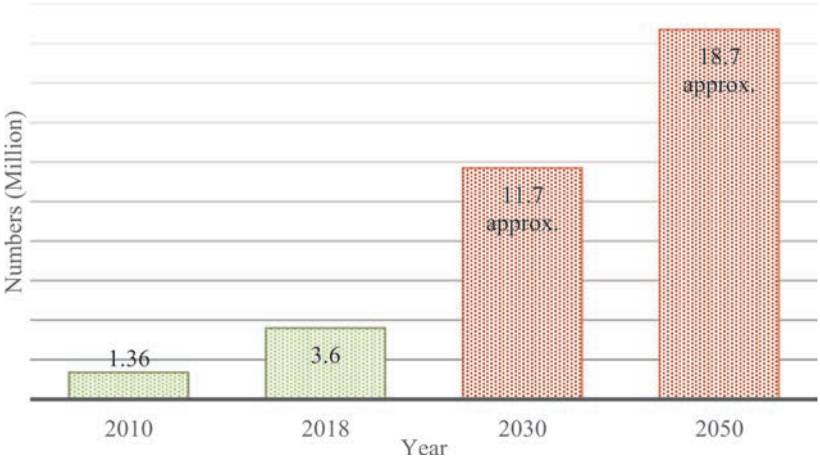

Figure 1.3 *Employment from investment in solar PV plants [1]*

power electronics switches to operate with pulse width modulation (PWM) makes the system-control efficient. Different control algorithms can be implemented for generating PWM and obtaining accurate output power with a low response time.

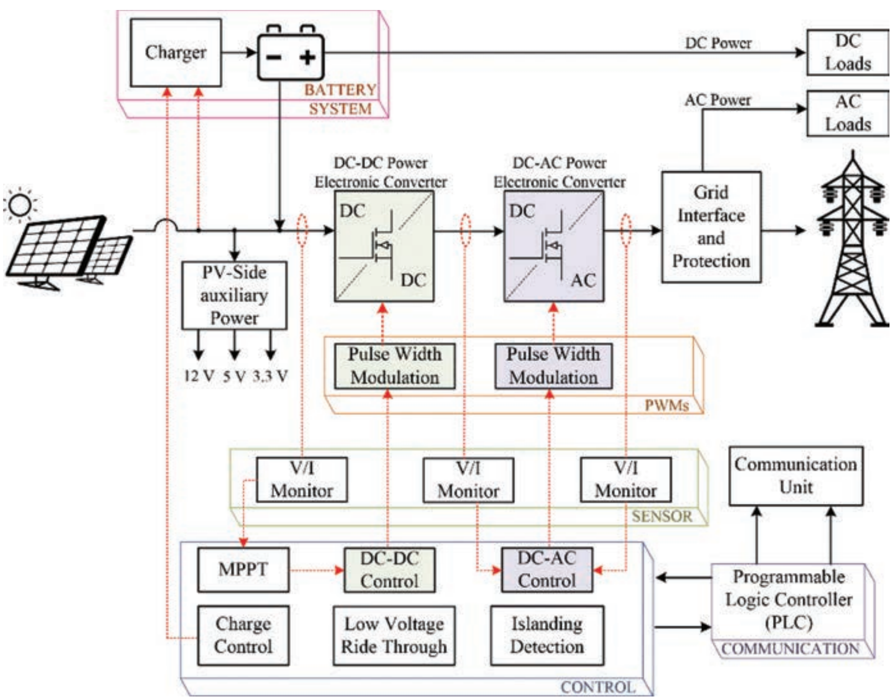

Figure 1.4 *Layout of solar PV plant*

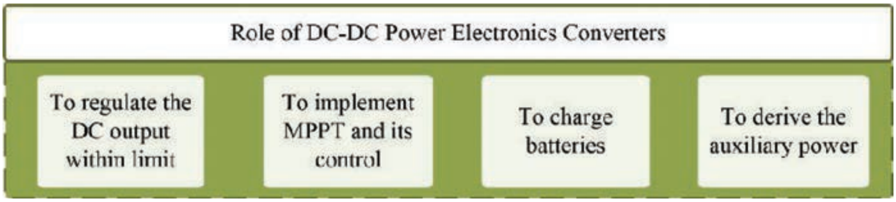

Figure 1.5 Main functions of DC–DC converters in solar PV plants

For a single-stage PV system, the output from the solar panel is directly fed to the DC–AC converter that regulated the incoming DC and converts it into AC. The output of the inverter is then passed through the filter before it is fed to the grid and local load. The control signal is generated by the measured voltage of the inverter at point of common coupling.

In the case of a two-stage system, the PV output from the solar panel is fed to the DC–DC converter. The maximum power point tracking (MPPT) control is implemented in the control of DC–DC converters. Also, if a battery is used in the solar PV plant, a charger circuit is required, which is also a power electronics converter. The PV-side auxiliary power is tapped off different DC voltage levels as per the requirement by the load. The main role of DC–DC converters in solar PV plant is shown in Figure 1.5. Similarly, the main functions of the DC–AC power electronics converter are shown in Figure 1.6. The LVRT and islanding detection protection are implemented in the control of the DC–AC converter.

1.3 DC–DC power electronics converters

In solar PV applications, DC–DC converters are used, and they are classified into two types: isolated and non-isolated type, as shown in Figure 1.7. The major difference between the two types of converters is in cost and galvanic isolation. The isolated type DC–DC has galvanic isolation between the input and output because of a magnetically coupled element, and at the same time, the cost is high. Flyback, resonant, forward, push-pull, bridge DC–DC converters are examples of isolated converters. The non-isolated type converters are Ćuk, SEPIC, boost, buck-boost, etc.

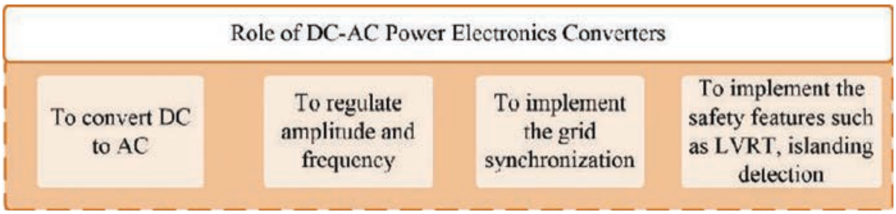

Figure 1.6 Main functions of DC–AC converters in solar PV plants

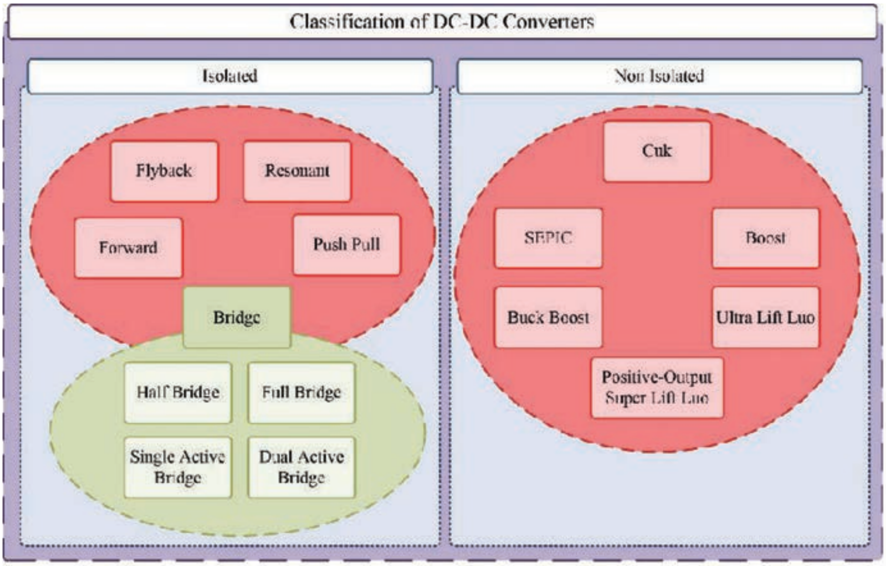

Figure 1.7 Classification of DC–DC converters used in solar PV applications

These DC–DC converters used in solar PV applications are discussed in Sections 1.3.1 to 1.3.14.

1.3.1 Buck converter

The buck converter, also known as a step-down converter, is used to step down the input voltage at the output side. The schematic of the converter is shown in Figure 1.8. This converter is a switched-mode power supply containing a diode, a power switch, and an energy storage element in the form of either a capacitor or a inductor. Here, the input voltage source feeds the controllable power switch

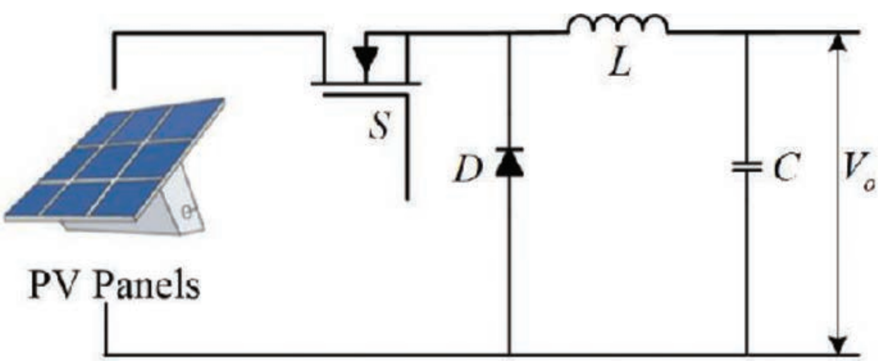

Figure 1.8 Schematic of a buck DC–DC converter

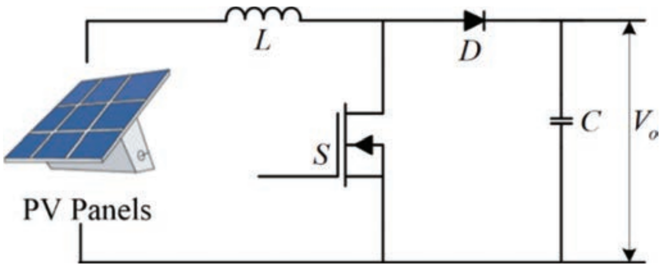

Figure 1.9 Schematic of the boost DC–DC converter

that is operated with a PWM either through a time base or with a frequency base. Generally, to eliminate the voltage ripple, a combination of capacitor and inductor can be used at both the load side and the supply side of the converter. The main application of this converter is in the battery charger circuit [7], solar PV pumping system [8], MPPT tracking, etc. [9].

1.3.2 Boost converter

The boost converter is referred to as a step-up converter and used in applications where the voltage magnitude at the output needs to be larger than the input voltage. This converter finds its majority of applications in PV systems to boost the PV voltage [10–12]. The schematic of the boost converter is shown in Figure 1.9. Similar to a buck converter, the boost converter is also a switched-mode power supply containing an inductor, a power switch, a diode, and a capacitor. Here, the input voltage source feeds the inductor that leads to a constant input current. Further, the power switch is operated with a PWM to achieve the required output voltage. Many modifications are available for the basic boost converter in the literature [10–13] to achieve ripple minimization, high voltage gain, and enhanced performance.

1.3.3 Buck-boost converter

The buck-boost converter can operate both in step-down and in step-up mode depending upon the duty cycle provided to the converter. The schematic of the converter shown in Figure 1.10 is developed by combining the basic buck converter

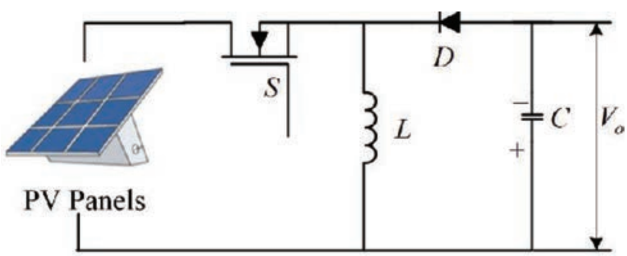

Figure 1.10 Schematic of the buck-boost DC–DC converter

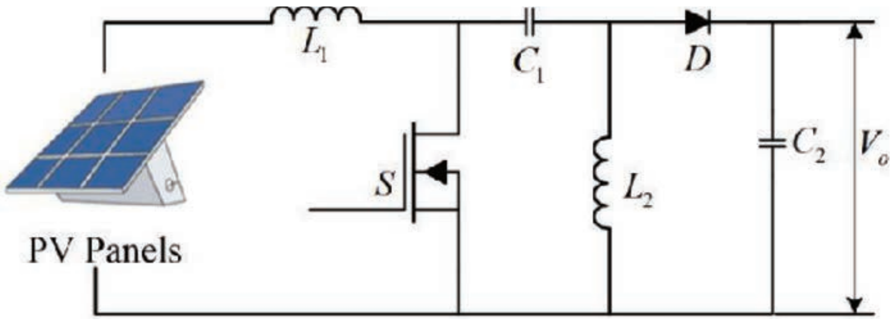

Figure 1.11 Schematic of the SEPIC DC–DC converter

and boost converter topologies discussed in Sections 1.3.1 and 1.3.2. This converter found most of its applications in stand-alone and grid-connected PV systems, and motor drives [14]. Similar to the operation of a buck converter, the input voltage source feeds the controllable power switch in the converter that is operated with a PWM. The literature identified that the continuous current mode operation of the buck-boost converter has lower ripples in the current. Further, a buck-boost converter with two power switches can have the least current and voltage stress on the components operating in the converter. Moreover, to enhance the operation of a basic buck-boost converter, various other topologies such as SEPIC [15], Ćuk [16], and Luo converters [17] are available in the literature.

1.3.4 Single-ended primary inductance converter

The single-ended primary inductance converter (SEPIC) is widely used in DC voltage flickering control and sensor-less control of PV applications. The schematic of the SEPIC is shown in Figure 1.11. While operating the converter, the on-time switching must be larger than the off-time switching to realize a high voltage at the output. Further, this also ensures that the capacitor is fully charged. For any condition, if the switching is not achieved, the converter fails to provide the required output. Besides, the operation of the converter along with a high-frequency transformer achieves an output voltage with minimized ripples. This provides various advantages with key features such as continuous output current, minimized output ripples, and minimized switching stress [15, 18–20].

1.3.5 Ćuk converter

The schematic of a Ćuk converter shown in Figure 1.12 has similarities with the basic buck-boost-converter except for the fact that the inductor is replaced with a capacitor to achieve the power transfer. This arrangement is also known as a negative-output capacitive energy-based flyback DC–DC converter [21]. Further, the Ćuk converter achieves ripple-free output in a system by inverting the output polarity of the converter with suitable connections [16, 22, 23]. Besides, to improve the efficiency of Ćuk converters and achieve optimal bidirectional operation concerning

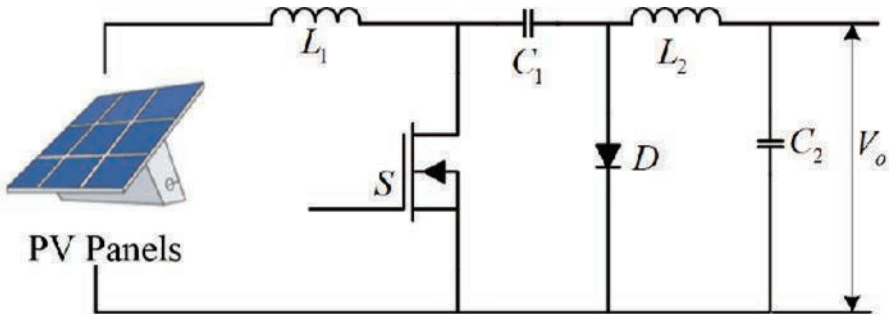

Figure 1.12 Schematic of a Ćuk DC–DC converter

the regulation of voltage and current [24], various modifications are proposed in the literature [25–28]. These studies have identified the application of the converter in various motor drive circuits [21] and renewable energy applications [29–31].

1.3.6 Positive-output super-lift Luo converter

The positive-output super-lift Luo converter shown in Figure 1.13 was initially introduced by Luo *et al*. [17] in 2003. This converter was developed with different series energy storage elements such as series inductors and capacitors that provide high output voltage resembling arithmetic progressions. Later, the design of the converter was modified in [32] by adding a high voltage transfer gain and its operation was enhanced by using a sliding mode controller in [33] for achieving the balance between the voltage regulation and load current. This converter is considered to be more powerful when compared to the SEPIC and Ćuk converters discussed in Sections 1.3.4 and 1.3.5 due to its unique features of enhanced efficiency and high

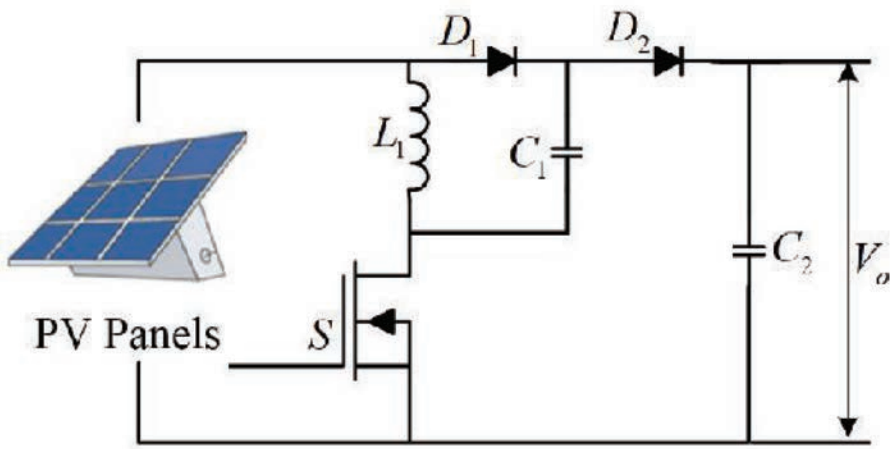

Figure 1.13 Schematic of a positive-output super-lift Luo DC–DC converter

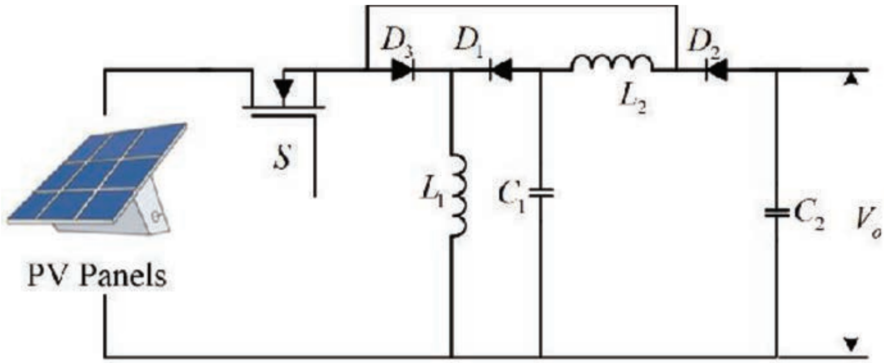

Figure 1.14 *Schematic of an ultra-lift Duo DC–DC converter*

output voltage resembling higher geometric progressions. Moreover, these converters are still under development for their operation with domestic and industrial PV applications [34, 35].

1.3.7 Ultra-lift Luo converter

The ultra-lift Luo converter is shown in Figure 1.14 [36, 37]. This converter combines the design aspects of voltage and super-lift Luo converters to produce a high-voltage conversion gain. This makes the converter highly efficient among the other non-isolated DC–DC converters. Further, it is identified that the closed-loop design of the converter is monotonous as the slightest variation in duty ratio results in large output voltage variations.

1.3.8 Zeta converter

The Zeta converter combines the advantages of the buck-boost, SEPIC, and Ćuk converters. The schematic of the Zeta converter is shown in Figure 1.15. When operated in a PV system, the Zeta converter enables continuous MPPT over the entire

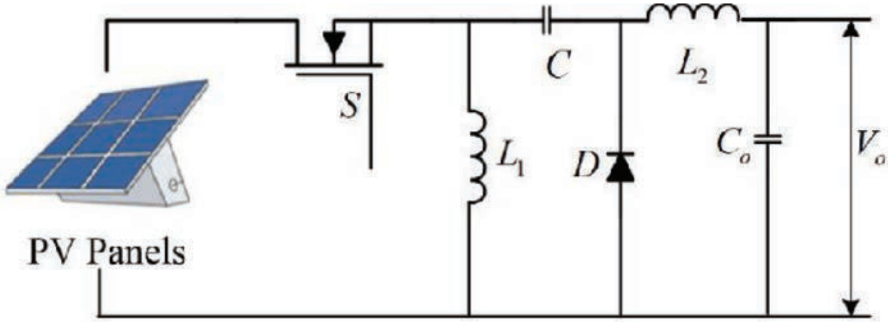

Figure 1.15 *Schematic of a Zeta DC–DC converter*

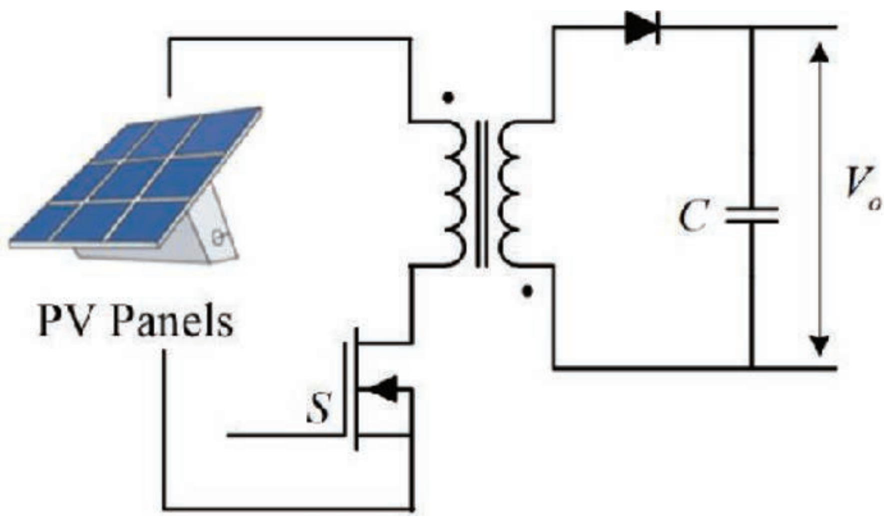

Figure 1.16 Schematic of a flyback DC–DC converter

area of the PV curve. Further, the Zeta converter provides a non-inverted output voltage that has either an enhanced or a diminished value concerning the input voltage [38–42]. Moreover, to reduce the output ripples and achieve enhanced voltage conversion in continuous and discontinuous modes, new topologies of the Zeta converter are developed in the literature [43]. These advancements are constituted for operation with the battery storage systems in PV applications.

1.3.9 Flyback converter

The schematic of the flyback DC–DC converter is shown in Figure 1.16. This converter acts as a key solution for higher converter gain requirements by employing transformers in the system. For a transformer with a large air gap to store energy, the flyback converter can be used in high-power applications. Further, the large air gap results in less magnetizing inductance, and the flyback converter provides very less energy transfer efficiency and large leakage flux. Moreover, the flyback converter overcomes the drawback of output polarity inversion and high current flow in the power switch and output diode in Ćuk converters [44]. These advantages have seen the application of flyback converters to operate in the discontinuous mode with isolated grid-connected inverters [45]. This application identified the unique features such as swift dynamic response and less complexity of the converter. Further, the efficiency and decreased ripple content of the converter is enhanced by employing various soft switching techniques [46, 47].

1.3.10 Three-port half-bridge DC–DC converter

The schematic of a three-port half-bridge DC–DC converter is shown in Figure 1.17. The converter's primary circuit operates in the buck converter mode

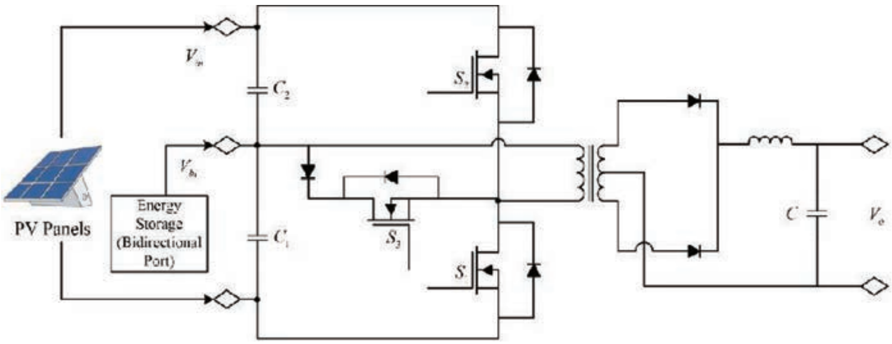

Figure 1.17 Schematic of a three-port half-bridge DC–DC converter

with synchronous rectification to provide the high-frequency transformer with a DC bias current. Further, various implementations are projected for post and synchronous regulations to regulate the three ports individually for achieving a single-stage power conversion with modest topology and simple control [48]. Moreover, to achieve continuous input current with a wide range of zero-voltage switching and low ripple, the three switches are operated with an active-clamped half-bridge DC converter [49]. Besides, to achieve a high voltage gain for large input voltage applications, the hybrid secondary rectifier is modified as a dual half-bridge LLC resonant converter. Here, depending on the switching strategy, the hybrid secondary rectifier acts as a quadruple rectifier [50]. In [51], the output power and efficiency are improved by employing an interleaved high-performance DC converter. A half-bridge of this topology ensures that the duty cycle of the two interleaved converters is close to 50 percent for achieving a continuous output current width with less lag and small component size. Further, to reduce the electromagnetic interference and voltage stress and achieve a high efficiency, the three-level converter is operated with a high-voltage bidirectional half-bridge in high-voltage DC microgrid applications [52].

1.3.11 Full-bridge converter

The full-bridge converter is used to integrate various components of the PV system such as PV array, energy storage device, and the load. The general schematic of a full-bridge converter is shown in Figure 1.18. The arrangement of a full-bridge converter consists of the integration of two buck-boost converters. This is developed to achieve zero-voltage switching and single power conversion with the topology [53]. Further, the circulating current losses are minimized by achieving zero-voltage switching during turn-on of switches through operating the full-bridge topology with asymmetrical PWM. Moreover, the use of asymmetrical PWM with a full-bridge converter minimizes the stress on power switches and achieves higher efficiency [54]. Besides, the problem of reverse recovery in output diode is overcome by achieving zero-current switchings during the switch turnoff through combining the resonant part of the circuit with the blocking capacitor and leakage inductance.

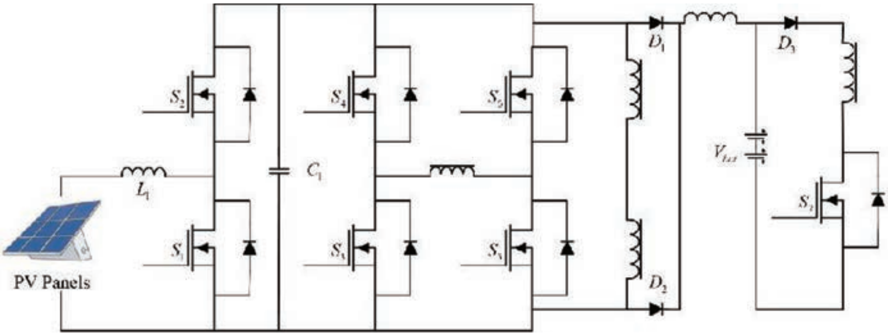

Figure 1.18 Schematic of a full-bridge DC–DC converter

1.3.12 Dual active bridge converter

The dual active bridge converter has found its application in stand-alone hybrid systems due to its advantages with high conversion efficiency, bidirectional power flow, galvanic isolation, and high power density [55–57]. The schematic of the converter is shown in Figure 1.19. From the circuit, it can be identified that the high-voltage DC sources feed the primary bridge and the low-voltage energy storage or load is connected to the secondary bridge. Further, a high-frequency power transformer is used to isolate the two full bridges whose leakage inductance is used as a storage element in the circuit. To enable the bidirectional power flow with the circuit, a square wave is conveniently phase shifted between both the bridges. Moreover, the voltage difference of the storage element is controlled to achieve power conversion with the circuit [58]. The control aspects of the dual active bridge-isolated bidirectional DC–DC converter are widely discussed through digital controllers in [59, 60]. In [61, 62], the high-frequency dual active bridge transformers were developed as an improvement to the existing converters. Besides, an ultra-capacitor-based dual active bridge converter was developed in [60].

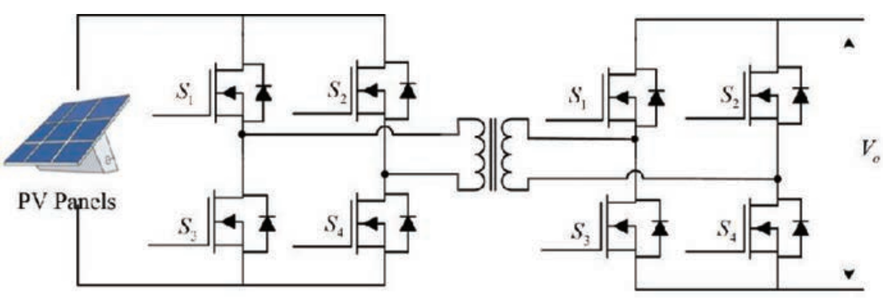

Figure 1.19 Schematic of a dual active bridge DC–DC converter

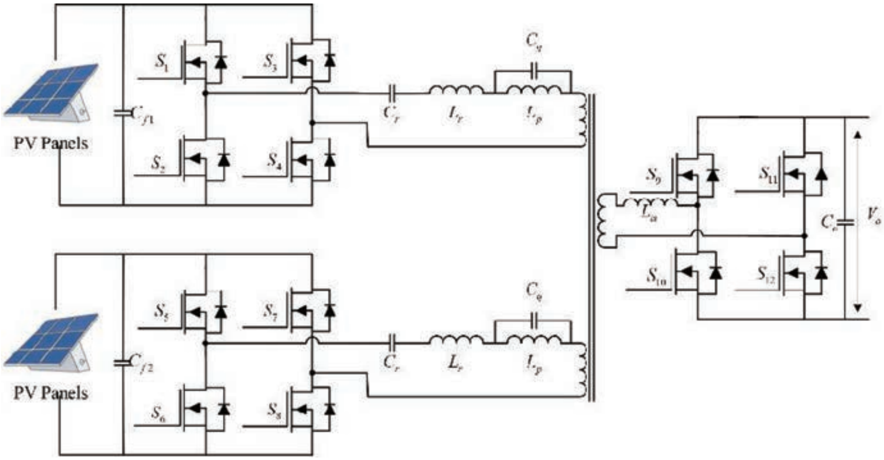

Figure 1.20 Schematic of a bidirectional multielement DC–DC resonant converter

1.3.13 Multielement resonant converter

The multielement resonant converter is an enhanced topology of the traditional LLC converter [63, 64]. This converter provides advantages of zero-voltage and -current switching using a short output circuit [65, 66], higher efficiencies at the full load operation, and high power density making them adaptable in renewable energy generation applications [67, 68]. The schematic of a three-port multielement resonant converter is shown in Figure 1.20. The design aspects of this converter involve series and parallel connection of five resonant components in the circuit. These multiple resonant components provide various resonant frequencies in the circuit and their suitable placement helps in transferring the active power of fundamental and third-order harmonics. Further, the parasitic leakage current due to the nonideal isolated transformer is ignored for this converter. From the literature, it is identified that the zero-voltage switching characteristics can be easily achieved for all the power switches in the three ports along with 96 percent power-conversion efficiency [69].

1.3.14 Push-pull converter

The push-pull converter, also known as a switching converter, is shown in Figure 1.21. This converter involves a transformer and operates with the help of a center-tapped primary winding by acting as a forward converter to the transformer core effectively. Further, the push-pull converter has small filters for different available power levels with the circuit. The major advantage of this converter is the transistor pair in the circuit employs input lines that avail the flow of current through the main winding of the transformer. Besides, the concurrent switching of the transistors draws current from the transformer resulting in a shattered condition for the current at the line during the switching condition of half-cycle pair. Moreover, when

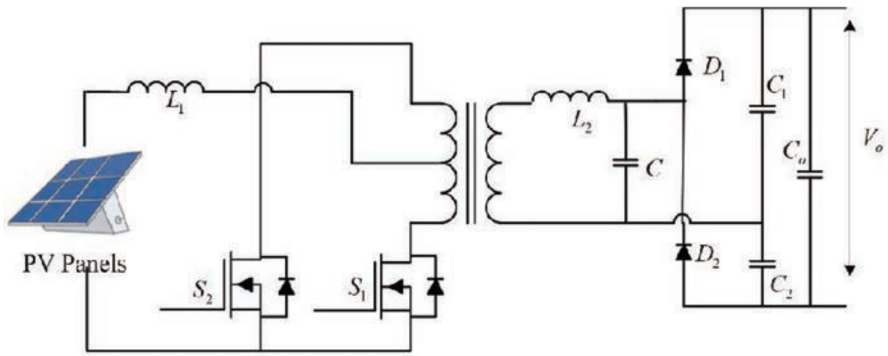

Figure 1.21 Schematic of a push-pull DC–DC converter

operated with low input noise, the push-pull converters have stable input current and can have efficient high-power applications [70]. Further, the center tapping that only utilizes half of the winding of the transformer at a time resulted in increased copper losses for the circuit.

Further, a summary of different DC–DC converters classified under isolated, non-isolated, unidirectional, and bidirectional are shown in Table 1.1 [71]. The summary identifies the important aspects of different converters and compares them with each other.

1.4 DC–AC converters

Solar inverters play a vital role in the PV application. The DC power from the panel after being boosted by a DC–DC converter is fed to the solar inverter before it can

Table 1.1 Comparison between different DC–DC converters

	Converter type	**Optimal power demand**	**Voltage stress**	**Efficiency**
Non-isolated unidirectional DC–DC converter	Buck	Low	High	Medium
	Boost	Low	High	Medium
	Buck-boost	Low	Medium	Medium
Non-isolated bidirectional DC–DC converter	Buck-boost	Low	Medium	Medium
	SEPIC	Low	High	Medium
	Ćuk	Low	High	Medium
	Half-bridge	Low	Medium	High
Isolated unidirectional DC–DC converter	Flyback	Low	High	High
	Half-bridge	Low	High	High
	Full-bridge	High	Medium	Medium
	Push-pull	High	High	Medium
Isolated bidirectional DC–DC converter	Half-bridge	Low	High	Medium
	Full-bridge	High	Low	Medium

be supplied to the load and grid. The inverter converts DC power into a regulated AC power that can be controlled by varying the switching of the power electronics devices. The basic inverter topologies are presented in Figure 1.22. The topologies are designed to operate in a grid-connected mode of operation while satisfying different grid codes requirements. From Figure 1.22, it can be deduced that the solar inverter can be categorized into two classes: transformer-based and transformerless inverters [72–74]. The galvanic isolation present in the transformer-based inverter provides isolation between the DC and the AC side and protects the system from any leakage power or power circulation that may occur in the AC side of the system. But the disadvantage of the transformer-based inverter is that the size of the inverter becomes large, and the efficiency of the inverter is low due to the transformer-based losses. To overcome the disadvantages, transformerless inverters were proposed. The absence of a transformer for the inverter reduces the size substantially and at the same time the efficiency of the inverter is improved [75]. However, the concern with transformerless inverters is the absence of isolation between the AC and DC side of the system. There is a change of power flow from AC to DC end, which may cause damage to DC equipment. As a result, many different topologies have been proposed to reduce leakage currents in transformerless inverters.

1.4.1 Transformer-based inverter

The transformer in the inverter provides galvanic isolation that separates the AC from the DC. The isolation avoids injection of DC current into the grid and stops leakage currents from the grid into the DC circuit. Transformer inverters can be categorized based on the operating frequency. The low-frequency transformer inverter as depicted in Figure 1.23a [76] is more reliable and cost-effective but at the same time, the efficiency is significantly poor. Whereas in the case of the high-frequency transformer inverter depicted in Figure 1.23b [77], the size of the inverter is small, and the inverter is more efficient compared to the low-frequency one. Because of the high-frequency operation, the components are constantly operating in high stress and the reliability is low [78, 79].

1.4.2 Transformerless inverter

The transformer present in the inverter achieves galvanic isolation and reduces the leakage currents from the grid. This helps in ensuring safety, but a substantial amount of power is lost because of the transformer, which reduces the efficiency of the inverter. Due to the large size and low efficiency of the transformer-based inverter, there has been a shift in research toward the transformerless inverters for PV applications. A few of the commonly used transformerless inverter topologies [80] are explained in this section. A comparative analysis of different transformerless inverters is presented in Table 1.2.

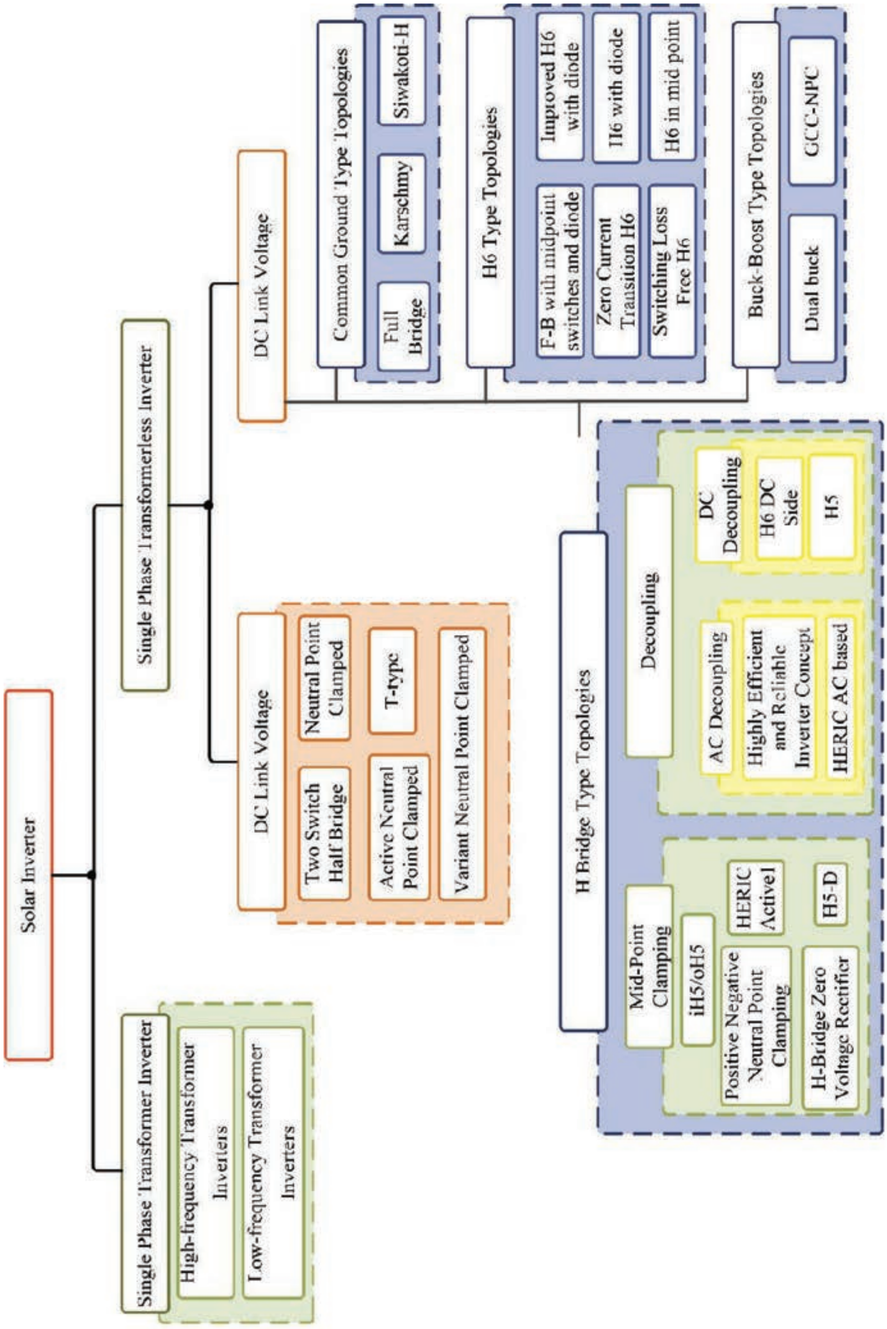

Figure 1.22 Classification of DC–AC converters used in solar PV application

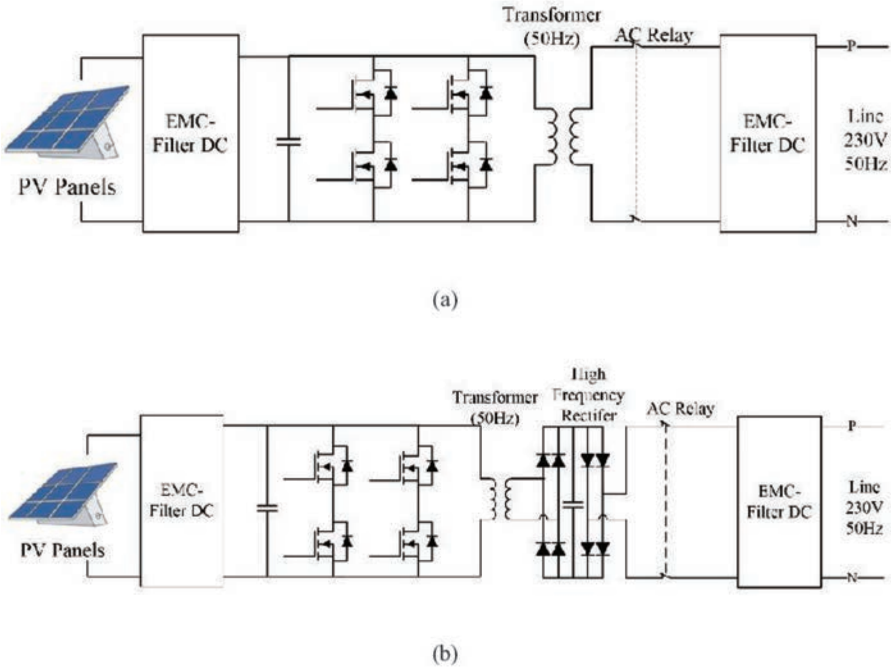

Figure 1.23 DC–AC converters with transformer: (a) low-frequency inverter, (b) high-frequency inverter

1.4.2.1 Half-bridge transformerless inverter

A half-bridge transformerless inverter comprises two power electronics switches connected in parallel with capacitors as shown in Figure 1.24 [105]. The operation is performed by turning on one switch at a time and charging the antiparallel capacitor at that instance. When the second switch is turned on, the corresponding capacitor is discharged, and the obtained inverter output is passed through an L-filter. The topology is simple to implement but achieving MPPT is difficult [106]. As a result, a high ripple is present in the output current.

1.4.2.2 Neutral point-clamped transformerless inverter

An neutral point-clamped (NPC) transformerless inverter comprises four power electronics switches with two diodes as shown in Figure 1.25 [82]. The clamping diodes at the midpoint aid in achieving the zero-voltage stage. Switches S_1 and S_3 operate in one-half cycle with alternating pulsing whereas S_2 and S_4 operate in another cycle. The current ripple is low compared to the half-bridge topology, but this topology is unable to balance conduction losses on the negative side causing a limitation for the DC-link [107].

Table 1.2 Comparison between different transformerless inverter topology

Single-phase transformerless inverter			Topologies	Components in topologies				Filter		Common-mode current	Common-mode voltage	Power factor (PF)	Total harmonic distortion (THD)	η (%)
				Semiconductor switches		Passive element								
				Insulated-gate bipolar transistor	Diodes	C	L	C	L					
DC-link voltage ($2V_{PV}$)			Two-switch H-B [81]	2	2	2	0	0	1	<20	Const.	N/A	1.1	N/A
			NPC [82]	4	2	2	0	0	1	= 0	Const.	N/A	1.19	N/A
			ANPC [83]	6	6	2	0	0	1	<20	N/A	N/A	N/A	N/A
			T-type [84]	4	4	2	0	0	1	<20	Const.	N/A	N/A	N/A
DC-link voltage (V_{PV})	Common-ground-type topologies		S4 [85]	4	2	3	0	1	1	=0	Const.	0.8	2.1	97.2
			Karschmy [86]	5	2	2	1	1	1	=0	Const.	Unity	N/A	N/A
			Siwakoti-H [87]	4	1	2	0	1	1	=0	Const.	0.85	<2.3	97.8
	H6-type topologies		Improved H6 with diode [88]	6	2	1	0	1	2	<20	190–200	0.9	N/A	96.5
			ZCT-H6 [89]	8	3	4	2	1	2	<250	N/A	Unity	N/A	95.6
			H6 with diode [90]	6	2	1	0	1	2	<200	159–240	0.99	1.86	97.3
			SLF-H6 [91]	8	2	4	2	1	2	<150	N/A	Unity	N/A	96.2
			H6 in midpoint [92]	6	0	1	0	1	2	<200	159–240	Unity	N/A	N/A
	Buck-boost type topologies		Dual buck [93]	4	2	2	3	1	2	N/A	N/A	N/A	3.6	98.3
			Generation control circuit-neutral point clamped [94]	6	2	2	1	1	1	<20	N/A	Unity	4.08	95.7
	H-bridge type topologies	Midpoint clamping	iH5/oH5 [95]	6	0	2	0	1	2	<20	199–200	Unity	N/A	96.9
			H5-D [96]	5	1	2	0	1	2	<50	185–195	Unity	4.88	95
			HERIC Active 1 [97]	7	2	2	0	1	2	<25	199–200	N/A	N/A	N/A
			HB-ZVR [98]	5	5	2	0	1	2	<200	163–200	Unity	N/A	94.8
			PN-NPC [99]	8	0	2	0	1	2	<35	199–201	Unity	N/A	97.2
		Decoupling	HERIC [100]	6	0	1	0	1	2	<200	165–235	Unity	N/A	97.1
			HERIC AC-based [101]	6	2	1	0	1	2	<200	165–236	Unity	N/A	N/A
			H6 DC side [102]	6	0	2	0	1	2	<200	151–249	Unity	1.58	95.9
			H5 [103, 104]	5	0	1	0	1	2	<200	159–235	Unity	N/A	98.5

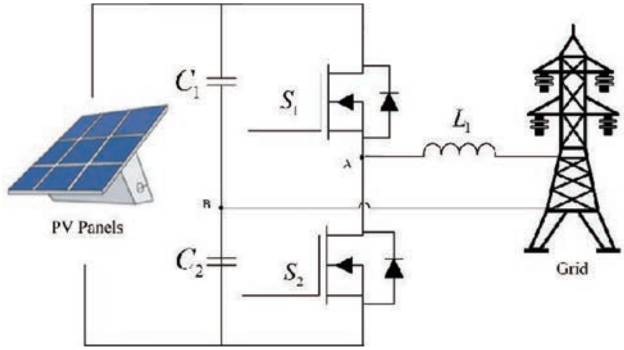

Figure 1.24 Two-switch half-bridge transformerless inverter

1.4.2.3 Active neutral point-clamped transformerless inverter

An active neutral point-clamped (ANPC) transformerless inverter illustrated in Figure 1.26 is a modification of an NPC inverter [108]. In the ANPC, the diodes are replaced by power electronics switches. The upper clamping is controlled by S_2 and S_5 whereas the lower clamping is regulated by S_3 and S_6 [109]. By replacing the diodes with the power electronics switches, the conduction losses of the inverter are controlled.

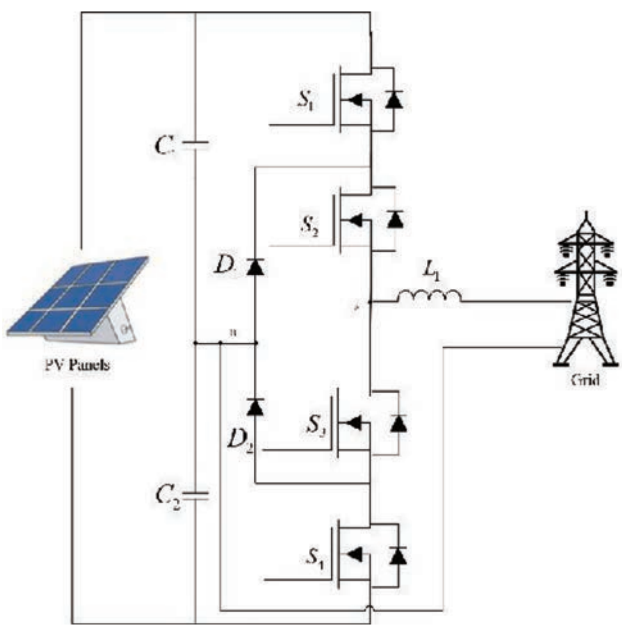

Figure 1.25 NPC transformerless inverter

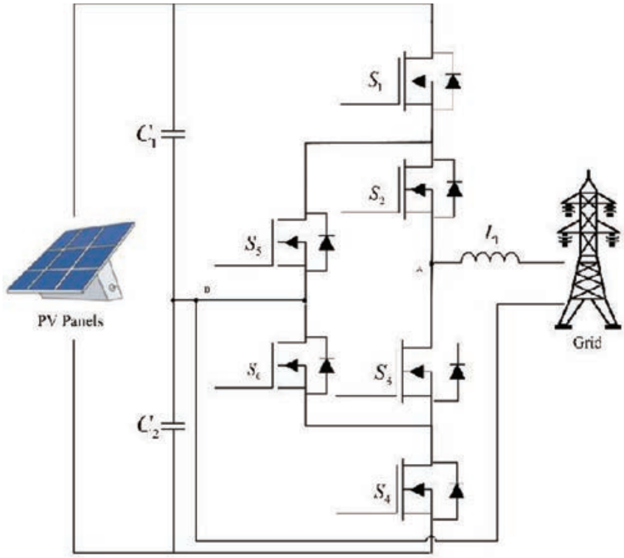

Figure 1.26 ANPC transformerless inverter

1.4.2.4 T-type transformerless inverter

A T-type three-level transformerless inverter consists of four power electronics switches with two bidirectional switches incorporated in the midpoint of the DC-link capacitor and switches (S_1 and S_2) leg [107] as illustrated in Figure 1.27. The S_1 and S_3 operate in a complementary way with S_2 and S_4 [110]. The clamping with two power electronics switches (S_3 and S_4) reduces the requirement of switching devices, which causes a reduction in conduction losses when compared to the ANPC topology.

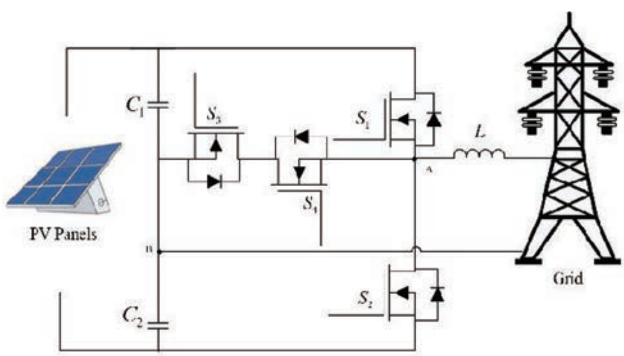

Figure 1.27 T-type transformerless inverter

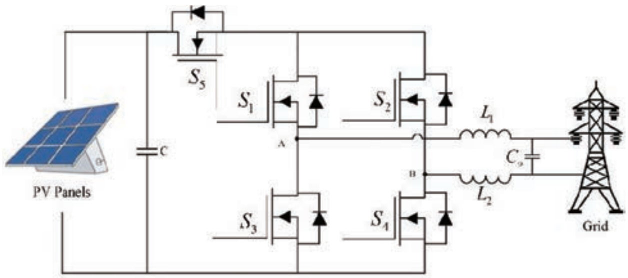

Figure 1.28 Three-switch transformerless inverter

1.4.2.5 Three-switch transformerless inverter

The three-switch transformerless inverter is a modified NPC or ANPC inverter. The number of power electronics switches is reduced from previous topologies, as shown in Figure 1.28. The topology incorporates a diode bridge along with S_3. The diode bridge provides the current path during the null state. The operation takes place in four modes. S_1 is turned on during the positive half-cycle whereas S_2 remains on during the negative half-cycle. During the freewheeling mode of a positive half-cycle, D_1 and D_4 are in forwarding bias with S_3 whereas for the negative half-cycle biasing, D_2 and D_3 are on alongside S_3 [111].

1.4.2.6 Full-bridge transformerless inverter

A full-bridge transformerless inverter consists of four power electronics switches as shown in Figure 1.29. During the positive half-cycle of operation, S_1 and S_4 are turned on and the antiparallel diode along S_2 and S_4 provide a path for the current flow. The S_2 and S_1 are complementary to each other and similarly S_3 and S_4 are complementary to each other. For positive half-cycle S_1 and S_4 are turned on and, as a result, the output voltage is equal to the input voltage. Whereas during the

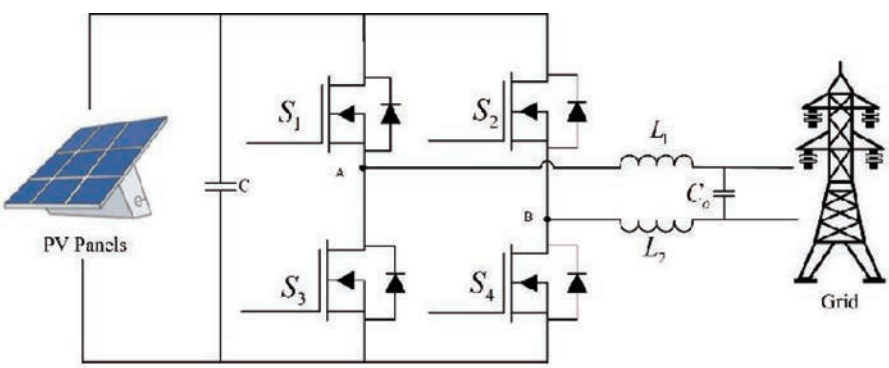

Figure 1.29 Full-bridge transformerless inverter

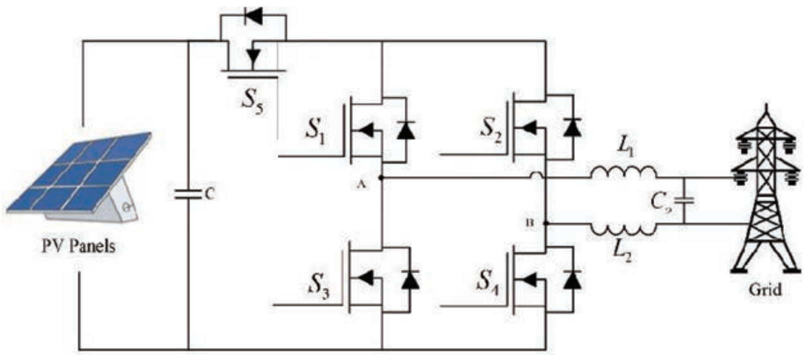

Figure 1.30 H5 transformerless inverter

freewheeling mode of the positive cycle, the current flows though S_1 and antiparallel diode of S_2 [112].

1.4.2.7 H5 transformerless inverter

The H5 transformerless inverter topology is the commonly implemented topology, which was patented and is commercially produced by SMA Solar Technology [103]. The topology consists of five power electronics switches as shown in Figure 1.30. It can be observed that S_5 on the DC side acts as a DC decoupling switch. During the freewheeling mode of operating, S_5 is turned off, which disconnects the DC side and effectively reduces the common-mode current [113, 114]. During the positive half-cycle, S_5 and S_4 operate at the switching frequency, whereas S_1 operates at the grid frequency. The other two switches remain in the off state. Whereas during the negative half-cycle, S_5 and S_2 operate at the switching frequency and S_3 operates at the grid frequency.

1.4.2.8 Full-bridge transformerless inverter with midpoint switches and diodes

The full-bridge transformerless inverter with midpoint switches and diodes topology is similar to the H5 transformerless inverter topology. Two extra switches are added on the top and bottom of a full-bridge inverter, as shown in Figure 1.31. While S_4 conduct, S_1 and S_6 conduct simultaneously. When S_3 is in on state, S_2 and S_5 operate with the same switching pulse. During a freewheeling period of a positive half-cycle, D_2 operates in forwarding bias along with S_4 . Whereas during the negative half-cycle, D_1 operates in forwarding bias along with S_2 [115].

1.4.2.9 H6-1 transformerless inverter

The H6-1 transformerless inverter consists of six power electronics switches, as shown in Figure 1.32 [116]. For operation in a positive half-cycle, S_1 , S_4, and S_6 are turned on. And during the freewheeling state, S_6 along with the antiparallel diode of

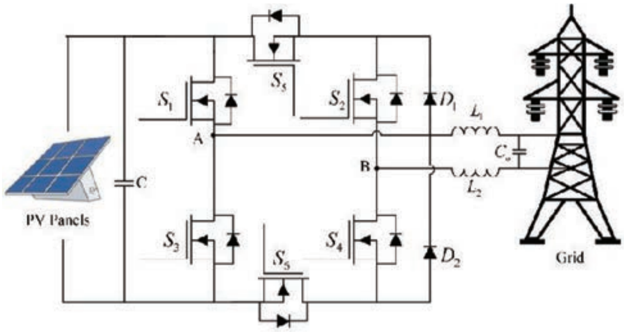

Figure 1.31 Full-bridge transformerless inverter with midpoint switches and diodes

S_5 conducts without any input, and the current flows through the load. During the negative half-cycle, S_2 , S_3, and S_5 are turned on and for freewheeling state, S_5 along with the antiparallel diode of S_6 conducts.

1.4.2.10 Modified Highly Efficient and Reliable Inverter Concept

The HERIC transformerless inverter consists of six power electronics switches and has an issue with leakage currents. To overcome the drawback a modified version was proposed in [117–119] as shown in Figure 1.33. The topology aims at reducing the leakage current and keeping the common-mode voltage to the minimum. The drawback of this topology remains the shoot-through issue.

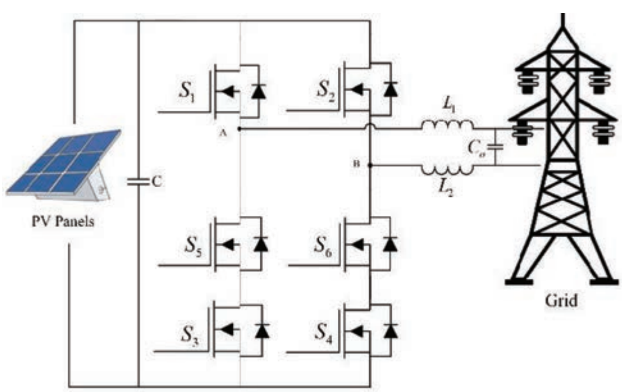

Figure 1.32 H6-1 transformerless inverter

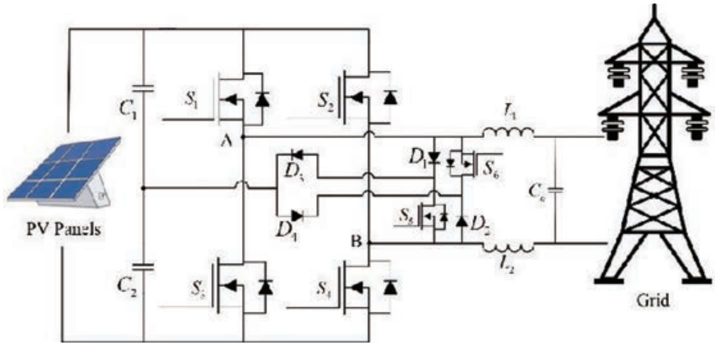

Figure 1.33 Modified HERIC transformerless inverter

1.4.2.11 oH5-1 transformerless inverter

The oH5-1 topology has been derived from the H5 transformerless inverter topology. The topology aims to clamp the input voltage to the half value [108]. The topology consists of six power electronics switches as shown in Figure 1.34.

1.4.2.12 Modified transformerless inverter

As illustrated in Figure 1.35, the topology presented is a modification over the basic full-bridge converter [120]. During the positive half-cycle S_1 and S_3 are in on state, whereas S_2 remains off. In case of the negative half-cycle, S_5 is in on state, whereas S_4 remains off.

1.5 Summary

Solar energy-based power generation is one of the major stakeholders in the renewable energy market around the globe. The low operating cost and easy availability

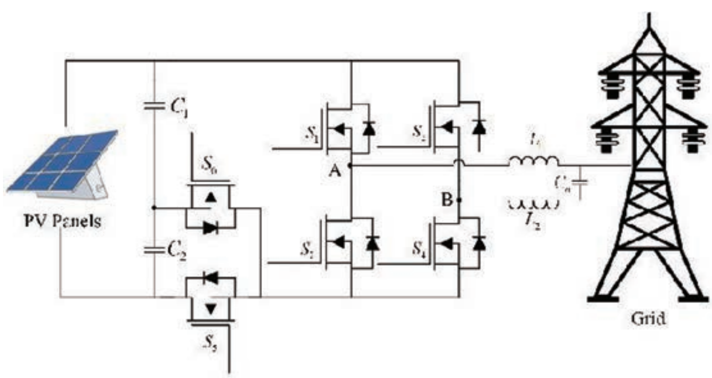

Figure 1.34 oH5-1 transformerless inverter

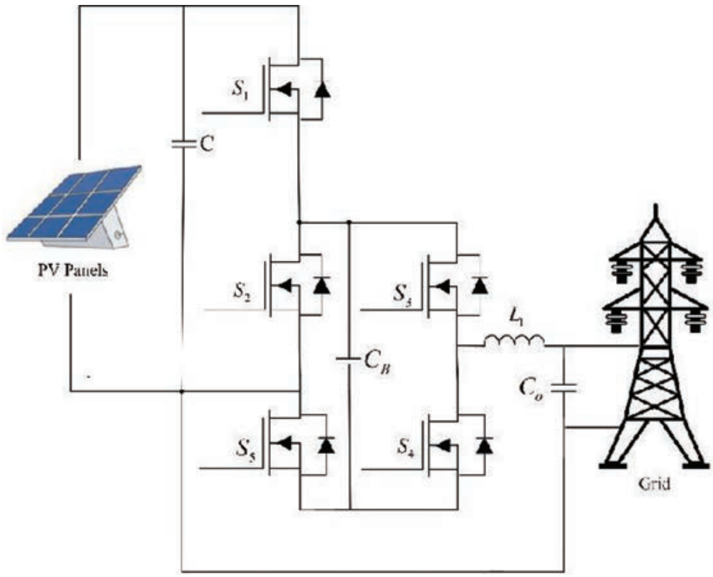

Figure 1.35 Modified transformerless inverter

have made it more favorable for the consumer. But many challenges are still present as a large amount of generated power is lost during the conversion process. As a result, many power electronics converters are designed over time to improve the efficiency and reliability of PV-based system. The chapter presents a brief introduction about the power electronics converters used for PV applications. Both the DC–DC converters and DC–AC converters are presented in this chapter and different topologies that are applied in PV applications are explained.

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